Extrema and the Mean Value Theorem

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Critical points and extrema**

Find the critical points of $f(x) = 2x^3 - 3x^2 - 12x + 5$.

$f'(x) = 6x^2 - 6x - 12 = 6(x^2 - x - 2) = 6(x - 2)(x + 1)$. Critical points: $x = 2$ and $x = -1$.

2 **[Foundational]** • **Target: Critical points and extrema**

Find the absolute maximum and minimum of $f(x) = x^3 - 3x$ on $[-2, 3]$, checking critical points and endpoints.

$f'(x) = 3x^2 - 3 = 0$ gives $x = \pm 1$ (both in $[-2, 3]$). Values: $f(-2) = -2$, $f(-1) = 2$, $f(1) = -2$, $f(3) = 18$. Absolute max: $f(3) = 18$. Absolute min: $f(-2) = f(1) = -2$ (tied).

3 **[Foundational]** • **Target: The Extreme Value Theorem**

State the Extreme Value Theorem and explain why it guarantees $f(x) = x^2 - 4x$ has an absolute max and min on $[0, 5]$.

The EVT states that a continuous function on a closed interval $[a, b]$ attains both an absolute maximum and an absolute minimum on that interval. Since $f(x) = x^2 - 4x$ is continuous (a polynomial) and $[0, 5]$ is closed, the EVT guarantees both extrema exist.

4 **[Foundational]** • **Target: Rolle's Theorem**

Verify that Rolle's Theorem applies to $f(x) = x^2 - 4x + 3$ on $[1, 3]$, and find the value(s) of c guaranteed by the theorem.

$f(1) = 0$ and $f(3) = 0$: equal endpoint values, and f is continuous and differentiable everywhere, so Rolle's Theorem applies. $f'(x) = 2x - 4 = 0$ gives $c = 2$, which lies in $(1, 3)$.

5 **[Foundational]** • **Target: Rolle's Theorem**

Explain why Rolle's Theorem does NOT apply to $f(x) = |x - 2|$ on $[0, 4]$, even though $f(0) = f(4) = 2$.

f is not differentiable at $x = 2$ (it has a corner), so one of Rolle's Theorem's hypotheses (differentiability on the open interval) fails — the theorem's guarantee does not apply.

6 **[Foundational]** • **Target: The Mean Value Theorem**

For $f(x) = x^3$ on $[0, 2]$, find the value(s) of c guaranteed by the Mean Value Theorem.

Average rate of change: $\frac{f(2) - f(0)}{2 - 0} = \frac{8}{2} = 4$. Need $f'(c) = 3c^2 = 4$: $c^2 = \frac{4}{3}$, $c = \frac{2}{\sqrt{3}} \approx 1.15$ (taking the positive root within $[0, 2]$).

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7 [Foundational] • Target: The Mean Value Theorem

A car travels 180 km in 2 hours. Using the Mean Value Theorem, explain why the car must have been traveling at exactly 90 km/h at some instant.

Average velocity $= \frac{180}{2} = 90$ km/h. Since position is a differentiable function of time, the MVT guarantees some instant c where the instantaneous velocity $v(c) = 90$ km/h exactly.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Using the MVT to bound function values

Given $f(2) = 5$ and $-3 \le f'(x) \le 4$ for all x, use the MVT to find the range of possible values for $f(6)$.

By the MVT, $f(6) - f(2) = f'(c)(6-2) = 4f'(c)$ for some $c \in (2, 6)$. Since $-3 \le f'(c) \le 4$: $-12 \le f(6) - 5 \le 16$, so $-7 \le f(6) \le 21$.

9 [Intermediate] • Target: Consequences of the MVT: monotonicity

Explain, using the MVT, why $f'(x) > 0$ on an interval implies f is strictly increasing on that interval.

For any $a < b$ in the interval, the MVT gives $\frac{f(b) - f(a)}{b - a} = f'(c)$ for some $c \in (a, b)$. Since $f'(c) > 0$ and $b - a > 0$, we get $f(b) - f(a) > 0$, i.e. $f(b) > f(a)$ — so f is strictly increasing.

10 [Intermediate] • Target: Consequences of the MVT: monotonicity

Explain why two functions f and g with $f'(x) = g'(x)$ for all x must differ by a constant, using the MVT.

Let $h(x) = f(x) - g(x)$. Then $h'(x) = f'(x) - g'(x) = 0$ for all x. If h were not constant, the MVT applied between any two points where h differs would give $h'(c) \neq 0$ for some c, a contradiction. So $h(x) = C$ for some constant C, i.e. $f(x) = g(x) + C$.

11 [Intermediate] • Target: Visualizing the Mean Value Theorem

The graph shows $f(x) = x^3$ on $[0, 2]$ together with the secant line through the endpoints and the tangent line at $x = \frac{2}{\sqrt{3}}$ (found earlier).

The tangent line at $c = \frac{2}{\sqrt{3}}$ is parallel to the secant line connecting $(0, 0)$ and $(2, 8)$, both having slope 4 — this is exactly the geometric statement of the MVT: somewhere between the endpoints, the instantaneous slope matches the average slope.

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12 **[Intermediate]** • **Target: Absolute extrema on a closed interval — a capstone problem**

Find the absolute maximum and minimum of $f(x) = x^4 - 4x^2 + 3$ on $[-2, 2]$, checking all critical points and endpoints.

$f'(x) = 4x^3 - 8x = 4x(x^2 - 2) = 0$ gives $x = 0, \pm\sqrt{2}$, all within $[-2, 2]$. Values: $f(-2) = 16 - 16 + 3 = 3$, $f(-\sqrt{2}) = 4 - 8 + 3 = -1$, $f(0) = 3$, $f(\sqrt{2}) = -1$, $f(2) = 3$. Absolute max: 3 (at $x = -2, 0, 2$). Absolute min: -1 (at $x = \pm\sqrt{2}$).

13 **[Intermediate]** • **Target: Rolle's Theorem with trigonometric functions**

Verify that Rolle's Theorem applies to $f(x) = \sin x$ on $[0, \pi]$, and find the value of c guaranteed by the theorem.

$f(0) = 0$ and $f(\pi) = 0$: equal endpoint values, and $\sin x$ is continuous and differentiable everywhere, so Rolle's Theorem applies. $f'(x) = \cos x = 0$ gives $c = \frac{\pi}{2}$, which lies in $(0, \pi)$.

14 **[Intermediate]** • **Target: An applied Mean Value Theorem problem**

A runner completes a 10 km race in 40 minutes. Using the Mean Value Theorem, find the runner's average speed, and explain why there must be some instant during the race when the runner's instantaneous speed equals this average.

Average speed $= \frac{10}{40} = 0.25$ km/min $= 15$ km/h (distance divided by time). Since distance is a differentiable function of time, the MVT guarantees some instant c during the race where the instantaneous speed equals exactly 15 km/h.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Using the MVT to bound function values** • **Cross-Lesson: 5.02 Concavity, Curve Sketching, and Optimization**

Level 2 · MediumFind the derivative $\frac{dy}{dx}$ for the function: $y = \int_{\cos x}^{x^3} \sqrt{1 - t^2}\, dt$.

Apply the Leibniz Rule with upper limit $u(x) = x^3 \Rightarrow u'(x) = 3x^2$ and lower limit $v(x) = \cos x \Rightarrow v'(x) = -\sin x$: $\frac{dy}{dx} = \sqrt{1 - (x^3)^2} \cdot (3x^2) - \sqrt{1 - (\cos x)^2} \cdot (-\sin x)$. Simplify trigonometric terms with $\sqrt{1 - \cos^2 x} = \sqrt{\sin^2 x} = |\sin x|$: $\frac{dy}{dx} = \mathbf{3x^2\sqrt{1 - x^6} + |\sin x|\sin x}$.

16 **[Challenging]** • **Target: Consequences of the MVT: monotonicity** • **Cross-Lesson: 6.01 Antiderivatives and Riemann Sums**

Level 1 · FoundationsFind the exact natural domain of the function $g(x) = \sqrt{\frac{x-1}{x+3}} - \ln(5 - x)$ in interval notation.

We must evaluate two simultaneous restrictions:Square root radicand must be non-negative and denominator non-zero: $\frac{x-1}{x+3} \geq 0 \quad \text{and} \quad x + 3 eq 0$. Sign analysis of the quotient with critical numbers $x = -3$ and $x = 1$: • For \(x Logarithm argument must be strictly positive: x

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17 **[Challenging]** • **Target: Visualizing the Mean Value Theorem** • **Cross-Lesson: 6.02 The Fundamental Theorem of Calculus**

Level 1 · FoundationsFind $h'(2)$ given the accumulation function $h(x) = \int_0^{x^2} \frac{t}{t+1}\,dt$.

By FTC Part 1 and the Chain Rule, let $u = x^2 \Rightarrow \frac{du}{dx} = 2x$: $h'(x) = \frac{u}{u+1} \cdot \frac{du}{dx} = \frac{x^2}{x^2+1} \cdot (2x) = \frac{2x^3}{x^2+1}$. Evaluate at $x = 2$: $h'(2) = \frac{2(2^3)}{2^2+1} = \frac{2(8)}{4+1} = \mathbf{\frac{16}{5}}$.

18 **[Challenging]** • **Target: Absolute extrema on a closed interval — a capstone problem** • **Cross-Lesson: 7.01 Differential Equations and Slope Fields**

Level 3 · Advanced Use the Divergence Theorem to find the outward flux of $\mathbf{F} = \langle x^3, y^3, z^3 \rangle$ across the boundary sphere $S : x^2 + y^2 + z^2 = 1$.

Compute the divergence: $abla \cdot \mathbf{F} = \frac{\partial}{\partial x}(x^3) + \frac{\partial}{\partial y}(y^3) + \frac{\partial}{\partial z}(z^3) = 3x^2 + 3y^2 + 3z^2 = 3(x^2 + y^2 + z^2)$ Apply the Divergence Theorem: $\iint_S \mathbf{F} \cdot d\mathbf{S} = \iiint_E 3(x^2 + y^2 + z^2)\,dV$ Convert to spherical coordinates: $x^2 + y^2 + z^2 = \rho^2$, $dV = \rho^2 \sin\phi\, d\rho\, d\phi\, d\theta$.

19 **[Challenging]** • **Target: Rolle's Theorem with trigonometric functions** • **Cross-Lesson: 8.01 Area Between Curves**

Level 2 · MediumEvaluate the trigonometric limit: $\lim_{x \to 0} \frac{1 - \cos(3x)}{x^2}$.

Multiply numerator and denominator by the conjugate $1 + \cos(3x)$: $\lim_{x \to 0} \frac{(1 - \cos(3x))(1 + \cos(3x))}{x^2(1 + \cos(3x))} = \lim_{x \to 0} \frac{1 - \cos^2(3x)}{x^2(1 + \cos(3x))} = \lim_{x \to 0} \frac{\sin^2(3x)}{x^2(1 + \cos(3x))}$. Re-group terms to leverage $\lim_{u \to 0} \frac{\sin u}{u} = 1$: $= \lim_{x \to 0} \left[\left(\frac{\sin(3x)}{3x} \right)^2 \cdot 3^2 \cdot \frac{1}{1 + \cos(3x)} \right]$. Evaluating individual limits.

20 **[Challenging]** • **Target: An applied Mean Value Theorem problem** • **Cross-Lesson: 8.02 Volumes by the Disc and Washer Methods**

Level 2 · MediumSolve the transcendental equation analytically for x in exact form using natural logarithms: $3^{2x+1} = 5^{x-2}$.

Take the natural logarithm of both sides: $\ln(3^{2x+1}) = \ln(5^{x-2})$. Apply the power law of logarithms: $(2x + 1)\ln 3 = (x - 2)\ln 5$. Distribute: $2x\ln 3 + \ln 3 = x\ln 5 - 2\ln 5$. Collect all terms containing x on one side: $2x\ln 3 - x\ln 5 = -\ln 3 - 2\ln 5 \Rightarrow x(2\ln 3 - \ln 5) = -(\ln 3 + 2\ln 5)$. Simplify logarithmic coefficients: $2\ln 3 = \ln 9$ and $2\ln 5 = \ln 25$: $x(\ln 9 - \ln 5) = -(\ln 3 + \ln 25) \Rightarrow x\ln\left(\frac{9}{5}\right) = -\ln(75)$. Solve for x.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Critical points and extrema

Given $f(-3) = -2$ and $-4 \le f'(x) \le 6$ for all x, use the MVT to find the largest possible value of $f(3)$.

a. -26

b. 40

c. 34 ✓

d. 28

By the MVT, $f(3) - f(-3) = f'(c)(6)$ for some $c \in (-3, 3)$. The largest value occurs when $f'(c) = 6$: $f(3) = -2 + 6(6) = 34$.

22 [Foundational] • Target: The Extreme Value Theorem

Find the larger critical point of $f(x) = x^3 - 9x^2 + 24x$.

a. 4 ✓

b. 2

c. 6

d. 8

$f'(x) = 3x^2 - 18x + 24 = 3(x - (2))(x - (4))$. Critical points: $x = 2$ and $x = 4$; the larger is 4.

23 [Foundational] • Target: Rolle's Theorem

Given $-3 \le f'(x) \le 2$ for all x, use the MVT to find the width of the range of possible values for $f(9) - f(2)$.

a. 31

b. 5

c. 42

d. 35 ✓

By the MVT, $f(9) - f(2) = f'(c)(7)$ ranges from $-3(7)$ to $2(7)$, a width of $(2 - (-3))7 = 5(7) = 35$.

24 [Foundational] • Target: The Mean Value Theorem

Does Rolle's Theorem guarantee a $c \in (-4, 6)$ with $f'(c) = 0$ for $f(x) = (x - 1)^2$ on $[-4, 6]$?

a. Yes, but only because f is increasing on the whole interval.

b. **Yes — $f(-4) = f(6) = 25$, and f is a polynomial (continuous and differentiable everywhere), so Rolle's Theorem applies; here $c = 1$.** ✓

c. No — although $f(-4) = f(6)$, f is not differentiable on $(-4, 6)$, so Rolle's Theorem does not apply.

d. No — Rolle's Theorem never applies to functions of the form $(x - 1)^2$.

Yes — $f(-4) = f(6) = 25$, and f is a polynomial (continuous and differentiable everywhere), so Rolle's Theorem applies; here $c = 1$.

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25 [Foundational] • Target: Using the MVT to bound function values

A continuous function on a closed, bounded interval $[a, b]$ is guaranteed (Extreme Value Theorem) to be:

a. bounded and to attain its max and min ✓

b. monotonic

c. unbounded

d. differentiable

The Extreme Value Theorem: continuity on a compact interval guarantees both boundedness and attained extrema.

26 [Foundational] • Target: Consequences of the MVT: monotonicity

Global extrema on closed interval: Extreme Value Theorem guarantees max and min are attained. Find them by checking:

a. **all critical points and endpoints** ✓

b. only critical points

c. only endpoints

d. only inflection points

EVT: critical points + endpoints.

27 [Foundational] • Target: Visualizing the Mean Value Theorem

Proof MVT: steps include Rolle's theorem applied to $h(x) = f(x) -$

a. $f'(a)(x - a)$

b. $f(a) - L(x)$ **where** L **is the secant line** ✓

c. $f(b)(x - a)$

d. $f(a)$

Define $h = f(x) - [f(a) + \frac{f(b) - f(a)}{b - a}(x - a)]$; apply Rolle.

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28 [Foundational] • Target: Absolute extrema on a closed interval — a capstone problem

The figure shows the graph of a continuous function $f(x)$ on the interval $[a, d]$. The shaded region represents the definite integral $\int_a^c f(x)\,dx$. If the area of the shaded region is 12 square units and the area of the unshaded region between $x = c$ and $x = d$ is 5 square units, what is the value of $\int_a^d f(x)\,dx$?

a. **17** ✓

b. 7

c. 2.4

d. -7

The Fundamental Theorem of Calculus relates the definite integral of a function to its antiderivative. Geometrically, the definite integral $\int_a^b f(x)\,dx$ represents the net signed area between the curve $f(x)$ and the x-axis from $x = a$ to $x = b$. Areas above the x-axis are positive, and areas below the x-axis are negative. The problem states that the area of the shaded region, representing $\int_a^c f(x)\,dx$, is 12 square units. This implies that this region contributes positively to the net signe.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Rolle's Theorem with trigonometric functions

The diagram shows the graph of the quadratic function $f(x) = ax^2 + bx - 2$. The graph passes through the points $(1, -3)$ and $(2, 2)$. Determine the value of a.

a. 1

b. **2** ✓

c. 3

d. 4

The function is given by $f(x) = ax^2 + bx - 2$. Since the graph passes through $(1, -3)$, we have $f(1) = -3$: $a(1)^2 + b(1) - 2 = -3$ $a + b - 2 = -3$ $a + b = -1$ (Equation 1) Since the graph passes through $(2, 2)$, we have $f(2) = 2$: $a(2)^2 + b(2) - 2 = 2$ $4a + 2b - 2 = 2$ $4a + 2b = 4$ $2a + b = 2$ (Equation 2) Now we solve the system of linear equations for a and b. Subtract Equation 1 from Equation 2: $(2a + b) - (a + b) = 2 - (-1)$ $a = 3$ Substitute $a = 3$ into Equation 1: $3 + b = -1$ $b = -4$ T.

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30 [Intermediate] • Target: An applied Mean Value Theorem problem

Consider the function $f(x) = x^2 - 3x + 2$. The graph of $y = f(x)$ is a parabola. Find the x-coordinates of the points where the graph intersects the x-axis.

a. 1 and 2 ✓

b. -1 and -2

c. 2 and 3

d. -1 and 2

To find the x-coordinates of the points where the graph intersects the x-axis, we need to find the values of x for which $f(x) = 0$. Set the function equal to zero: $x^2 - 3x + 2 = 0$ Factor the quadratic equation: $(x - 1)(x - 2) = 0$ This gives two possible solutions for x: $x - 1 = 0 \Rightarrow x = 1$ $x - 2 = 0 \Rightarrow x = 2$ Therefore, the graph intersects the x-axis at $x = 1$ and $x = 2$.

31 [Intermediate] • Target: Critical points and extrema

The graph of the function $f(x) = x^2 - 4x + 3$ is shown below. The graph intersects the x-axis at points A and B. The vertex of the parabola is at point C. Determine the coordinates of the point where the graph intersects the y-axis.

a. (0, 3) ✓

b. (3, 0)

c. (0, -1)

d. (-1, 0)

To find the y-intercept of the graph of a function, we set $x = 0$ in the function's equation. For the function $f(x) = x^2 - 4x + 3$, substituting $x = 0$ gives: $f(0) = (0)^2 - 4(0) + 3$ $f(0) = 0 - 0 + 3$ $f(0) = 3$ Therefore, the graph intersects the y-axis at the point (0, 3).

32 [Intermediate] • Target: The Extreme Value Theorem

Consider the function $f(x) = \frac{ax^2 + bx + c}{dx + e}$. The graph of $y = f(x)$ has a vertical asymptote at $x = 2$ and a horizontal asymptote at $y = 3$. The graph also passes through the points $(0, 0)$ and $(1, -3)$. Find the value of $\frac{a}{d}$.

a. -6

b. -3

c. 3 ✓

d. 6

The vertical asymptote at $x = 2$ implies that the denominator $dx + e$ is zero when $x = 2$, so $2d + e = 0$, which means $e = -2d$. Thus, the denominator can be written as $d(x - 2)$. The function becomes $f(x) = \frac{ax^2 + bx + c}{d(x - 2)}$. The horizontal asymptote at $y = 3$ implies that the limit of $f(x)$ as $x \to \pm\infty$ is 3. For a rational function, this limit is the ratio of the leading coefficients if the degree of the numerator is equal to the degree of the denominator. Here, the degree.

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33 [Intermediate] • Target: Rolle's Theorem

Consider the function $f(x) = \frac{ax^2 + bx + c}{x - d}$, where $a, b, c, d \in \mathbb{R}$. The graph of $y = f(x)$ has a vertical asymptote at $x = 3$ and a horizontal asymptote at $y = 2$. The function passes through the point $(1, -8)$. If $f'(x)$ has a critical point at $x = 5$, find the value of d.

a. 1

b. 2

c. **3** ✓

d. 4

The vertical asymptote at $x = 3$ implies that the denominator is zero at $x = 3$, so $d = 3$. The horizontal asymptote $y = 2$ for a rational function where the degree of the numerator is equal to the degree of the denominator is the ratio of the leading coefficients. Therefore, $\frac{a}{1} = 2$, which means $a = 2$. Now we have $f(x) = \frac{2x^2 + bx + c}{x - 3}$. The function passes through $(1, -8)$: $-8 = \frac{2(1)^2 + b(1) + c}{1 - 3} = \frac{2 + b + c}{-2}$ $16 = 2 + b + c \Rightarrow b + c = 14$. Next, we find the.

34 [Intermediate] • Target: The Mean Value Theorem

If $|f^{(4+1)}(c)| \le 1$ on the relevant interval, bound the Lagrange error $|R_4(x)|$ for approximating f at $x = 0.1$ using a Taylor polynomial centred at $a = 1$.

a. 0.0098415

b. **0.00492075** ✓

c. 0.00246038

d. 0.0273375

Lagrange error bound: $|R_4(x)| \le \frac{M|x - a|^{4+1}}{(4+1)!} = \frac{1 \cdot 0.9^5}{5!} \approx 0.00492075$.

35 [Intermediate] • Target: Using the MVT to bound function values

Find the flux of $\mathbf{F} = \langle x, y, z \rangle$ through the closed unit sphere using the Divergence Theorem.

a. 2π

b. $\frac{4\pi}{3}$

c. 4π ✓

d. $\frac{8\pi}{3}$

$abla \cdot \mathbf{F} = 3$. Volume of unit sphere $= \frac{4\pi}{3}$. Flux $= 3 \cdot \frac{4\pi}{3} = 4\pi$.

36 [Intermediate] • Target: Consequences of the MVT: monotonicity

Local extrema of f can occur only at critical points (interior) or on the:

a. axes

b. level curves

c. **boundary** ✓

d. Hessian eigenvalues

Global extrema on closed bounded region: check interior critical points and boundary.

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37 [Intermediate] • Target: Visualizing the Mean Value Theorem

The divergence theorem applied to $\mathbf{F} = (x^3, y^3, z^3)$ over unit ball gives $\oiint_S \mathbf{F} \cdot \hat{n}\, dS = \iiint_V 3(x^2 + y^2 + z^2) dV =$

a. $4\pi/3$

b. 3π

c. $12\pi/5$ ✓

d. π

$\int_0^1 3r^2 \cdot 4\pi r^2 dr = 12\pi \int_0^1 r^4 dr = 12\pi/5.$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Absolute extrema on a closed interval — a capstone problem • Cross-Lesson: 7.01 Differential Equations and Slope Fields

The method of Lagrange multipliers finds constrained extrema using multiplier λ. The critical value of λ itself:

a. **is the rate of change of the optimal value w.r.t. the constraint constant** ✓

b. gives the extremum value

c. is always zero

d. equals $|ablaf|$

The Lagrange multiplier = sensitivity (shadow price).

39 [Challenging] • Target: Rolle's Theorem with trigonometric functions • Cross-Lesson: 8.01 Area Between Curves

Mean Value Theorem guarantees c in open interval — this means the point is:

a. an endpoint

b. **strictly interior** ✓

c. a maximum

d. undefined

Open interval (a, b): strictly interior.

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40 [Challenging] • Target: An applied Mean Value Theorem problem • Cross-Lesson: 8.02 Volumes by the Disc and Washer Methods

Closed interval method: candidates for abs extrema are:

a. only critical points

b. only endpoints

c. **critical points and endpoints ✓**

d. inflection points

All critical points interior + endpoints.

41 [Challenging] • Target: Critical points and extrema • Cross-Lesson: 1.01 Limits Graphically and Numerically

Analytic function: uniqueness theorem — if f, g analytic on connected open set, and $f = g$ on a sequence of points converging to an interior point, then $f = g$:

a. on the boundary only

b. at all integers

c. at a dense subset

d. **everywhere on the domain ✓**

Identity theorem for analytic functions.

42 [Challenging] • Target: The Extreme Value Theorem • Cross-Lesson: 1.02 Continuity and the Intermediate Value Theorem

Consider the quadratic function $f(x) = x^2 - 8x + k$. The graph of $y = f(x)$ is a parabola that intersects the x-axis at the points $(2, 0)$ and $(6, 0)$. Find the value of k.

a. **12 ✓**

b. 16

c. 20

d. 24

Since the parabola intersects the x-axis at $(2, 0)$ and $(6, 0)$, these are the roots of the quadratic equation $f(x) = 0$. Therefore, we can write the function in factored form as $f(x) = a(x - 2)(x - 6)$. Expanding this, we get $f(x) = a(x^2 - 8x + 12)$. Comparing this with the given function $f(x) = x^2 - 8x + k$, we can see that the coefficient of x^2 is 1. Thus, $a = 1$. So, $f(x) = x^2 - 8x + 12$. By comparing the constant terms, we find that $k = 12$. Alternatively, we can substitute one of the roots into the o.

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43 [Challenging] • Target: Rolle's Theorem • Cross-Lesson: 2.01 The Definition of the Derivative

The diagram shows the graph of the function $f(x) = x^2 - 4x + 3$ and a line $y = g(x)$. The line $y = g(x)$ intersects the parabola $y = f(x)$ at points A and B. Point A is given as $(1, 0)$. Find the x-coordinate of point B.

a. **3** ✓

b. 4

c. 5

d. 2

The points of intersection A and B are found by setting the two functions equal to each other: $f(x) = g(x)$. We are given $f(x) = x^2 - 4x + 3$ and from the diagram, the line is $y = x - 1$, so $g(x) = x - 1$. Setting them equal: $x^2 - 4x + 3 = x - 1$. Rearranging the equation to form a quadratic equation: $x^2 - 4x - x + 3 + 1 = 0$, which simplifies to $x^2 - 5x + 4 = 0$. We can factor this quadratic equation: $(x - 1)(x - 4) = 0$. The solutions for x are $x = 1$ and $x = 4$. We are given that point A has an x-c.

44 [Challenging] • Target: The Mean Value Theorem • Cross-Lesson: 2.02 Product, Quotient, and Trigonometric Derivatives

The diagram shows the graph of the function $y = f(x)$. The function is defined by $f(x) = x^2 + bx + c$. The graph passes through the point $(1, 0)$ and has a vertex at $(2, -1)$. Find the value of b.

a. **-4** ✓

b. 4

c. -2

d. 2

The vertex of a parabola given by the equation $y = ax^2 + bx + c$ is at $x = -\frac{b}{2a}$. In this case, $a = 1$. The vertex is given as $(2, -1)$, so the x-coordinate of the vertex is 2. Therefore, $2 = -\frac{b}{2(1)}$, which implies $2 = -\frac{b}{2}$. Multiplying both sides by 2 gives $4 = -b$, so $b = -4$. Alternatively, using the vertex form of the quadratic equation, $y = a(x - h)^2 + k$, where (h, k) is the vertex. Here, $a = 1$, $h = 2$, and $k = -1$. So, $y = 1(x - 2)^2 - 1$. Expanding this, we get.

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45 [Challenging] • Target: Using the MVT to bound function values • Cross-Lesson: 3.01 The Chain Rule

Consider the function $f(x) = x^2 - 5x + 6$. The graph of this function is a parabola. Determine the x-coordinates of the points where the graph intersects the x-axis.

a. $1, 6$

b. $2, 3$ ✓

c. $1, 2$

d. $3, 4$

To find the x-coordinates where the graph intersects the x-axis, we need to find the roots of the function, i.e., where $f(x) = 0$. We set the function equal to zero: $x^2 - 5x + 6 = 0$. This is a quadratic equation that can be factored. We are looking for two numbers that multiply to 6 and add to -5. These numbers are -2 and -3. So, the equation can be factored as: $(x - 2)(x - 3) = 0$. For the product of two factors to be zero, at least one of the factors must be zero. Therefore, $x - 2 = 0$ or $x - 3 = 0$. Solvi.

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